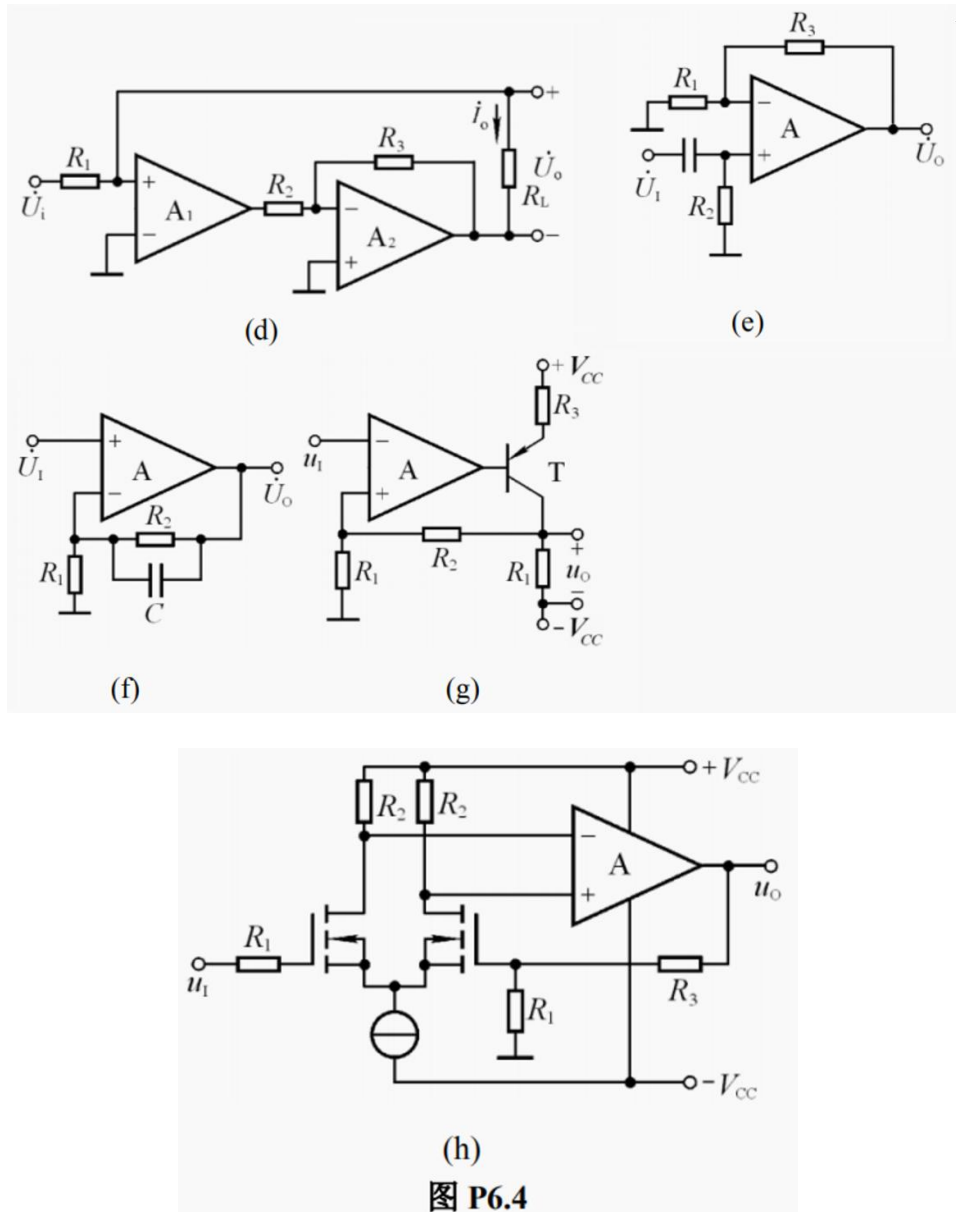


## Homework 7

✧ Strong negative feedback & Self-excited oscillation: 6.6/6.8/ 6.9/ 6.10

**6.6** 分别判断图 6.4 (d)~(h)所示各电路中引入了哪种组态的交流负反馈（交流负反馈组态为上次作业内容），并计算它们的反馈系数。

6.6 Determine the type of AC negative feedback in the amplifier circuits shown in Fig. 6.4 (d) - (h) (this is the question from the last homework), and calculate their feedback factors.



**6.8** 分别估算图 6.4 (d)~(h)所示各电路在理想运放条件下的电压放大倍数。

6.8 Assume all integrated operational amplifiers in Fig. 6.4 (d) - (h) are ideal, please calculate the voltage gain of each circuit.

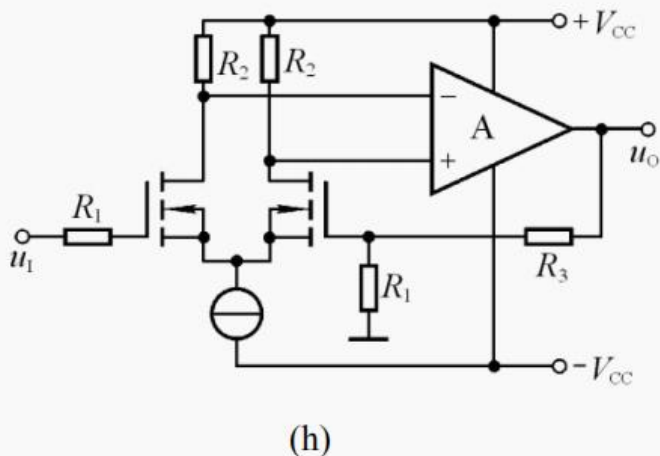
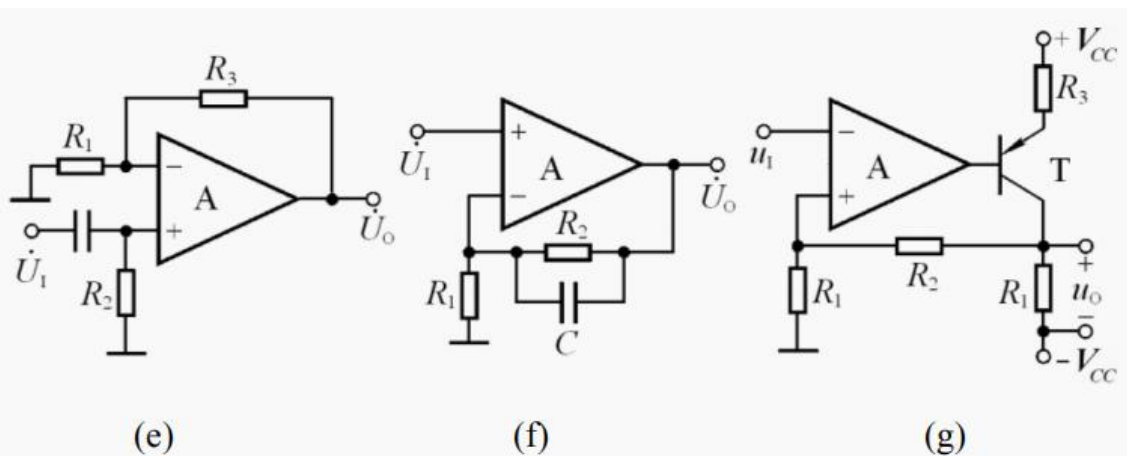
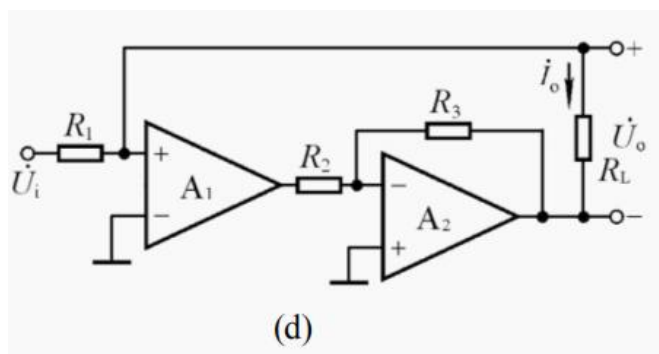
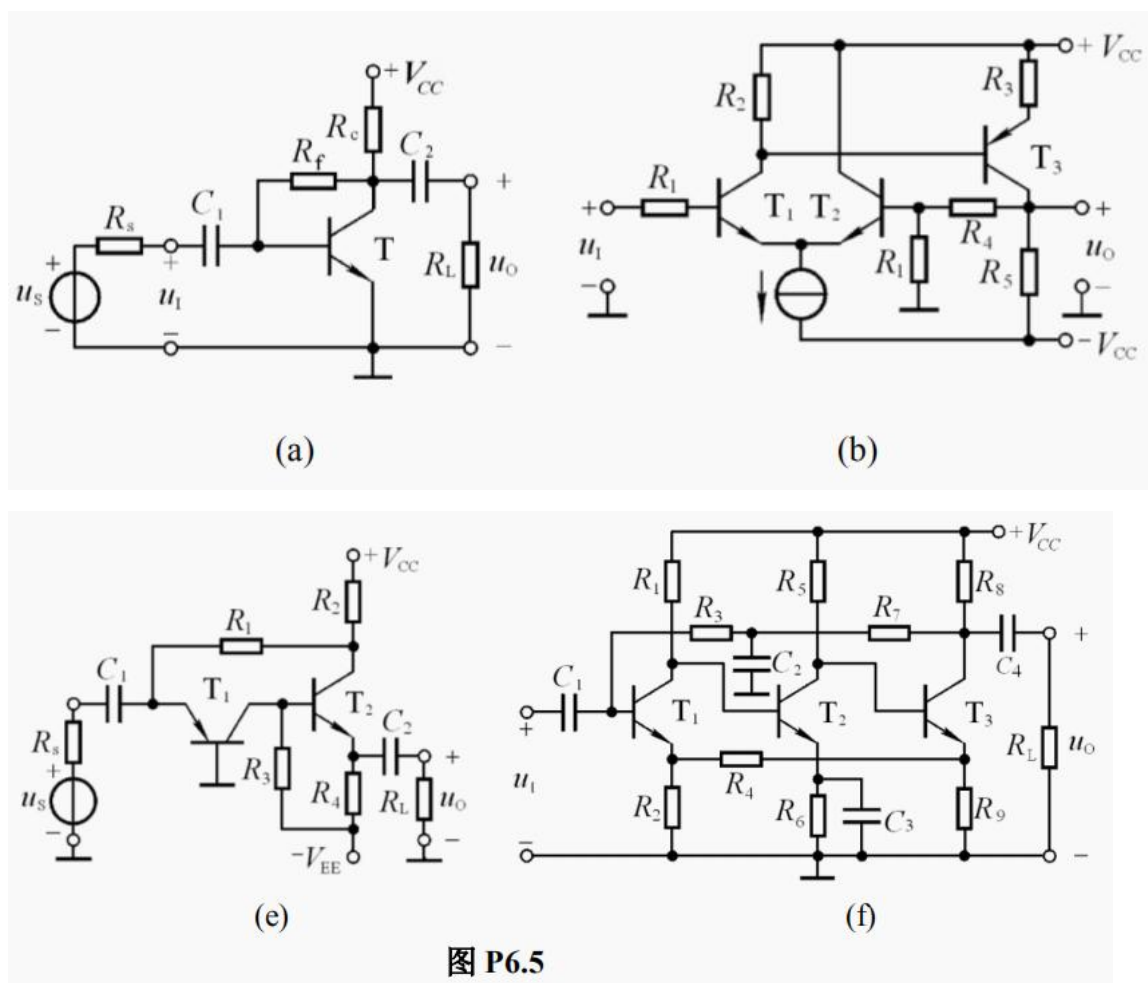
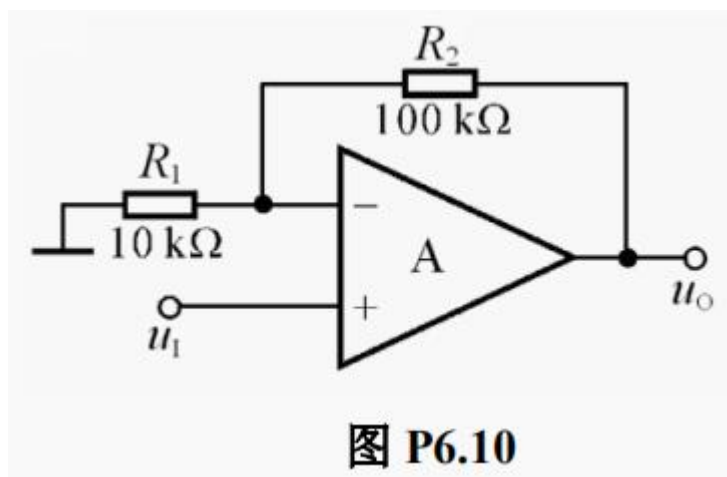


图 P6.4

6.9 分别估算图 6.5 (a)、(b)、(e)、(f)所示各电路在深度负反馈条件下的电压放大倍数。  
 6.9 Please estimate the voltage gain of each circuit shown in Fig. 6.5 (a), (b), (e) and (f) under the condition of strong negative feedback.



6.10 电路如图 P6.10 所示，已知集成运放为理想运放，最大输出电压幅值为  $\pm 14V$ 。



填空：

电路引入了\_\_\_\_\_（填入反馈组态）交流负反馈，电路的输入电阻趋近于\_\_\_\_\_，

电压放大倍数  $A_{uf} = \Delta u_O / \Delta u_I \approx$  \_\_\_\_\_。

设  $u_i = 1V$ ，则  $u_o =$  \_\_\_\_\_  $V$ ；

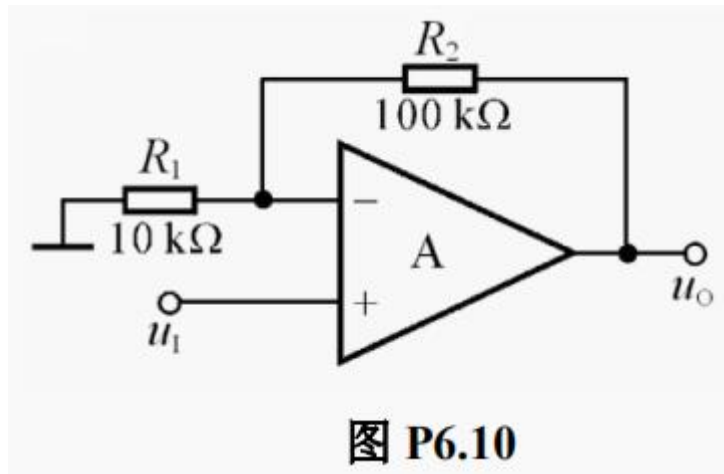
若  $R_1$  开路, 则  $u_o$  变为 \_\_\_\_\_  $V$  ;

若  $R_1$  短路, 则  $u_o$  变为 \_\_\_\_\_  $V$  ;

若  $R_2$  开路, 则  $u_o$  变为 \_\_\_\_\_  $V$  ;

若  $R_2$  短路, 则  $u_o$  变为 \_\_\_\_\_  $V$  。

6.10 Assume all integrated operational amplifiers are ideal, and the maximum output voltage is  $U_{om} = \pm 14V$



(1) Judge the type of AC negative feedback of the circuit shown In figure P6.10, (Voltage-series, Voltage-parallel, Current-series, Current-parallel) ?

(2) Input resistance  $R_i = ?$

(3) Voltage gain  $A_{uf} = \Delta u_o / \Delta u_i \approx ?$

(4) If  $U_i = 1V$ ,  $U_o =$  \_\_\_\_\_  $V$

If  $R_1 = \infty$ ,  $U_o =$  \_\_\_\_\_  $V$

If  $R_1 = 0$ ,  $U_o =$  \_\_\_\_\_  $V$

If  $R_2 = \infty$ ,  $U_o =$  \_\_\_\_\_  $V$

If  $R_2 = 0$ ,  $U_o =$  \_\_\_\_\_  $V$